Home Tasks - 05

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Q1.

Julia asked her students to create some coding challenges. Write a query to print the hacker_id, name, and the total number of challenges created by each student. Sort your results by the total number of challenges in descending order. If more than one student created the same number of challenges, then sort the result by hacker_id. If more than one student created the same number of challenges and the count is less than the maximum number of challenges created, then exclude those students from the result.

Input Format

The following tables contain challenge data:

Column	Type
hacker_id	Integer
name	String

- Hackers: The hacker_id is the id of the hacker, and name is the name of the hacker.
- Challenges: The challenge_id is the id of the challenge, and hacker_id is the id of the student who created the challenge.

Column	Type
challenge_id	Integer
hacker_id	Integer

Solution:

```
SELECT H.HACKER_ID, H.NAME, COUNT(*) AS CHALLENGES_CREATED
FROM HACKERS H
JOIN CHALLENGES C
ON H.HACKER_ID = C.HACKER_ID
GROUP BY H.HACKER_ID, H.NAME
HAVING COUNT(*) = (
    SELECT MAX(CHALLENGES_COUNT)
    FROM (
        SELECT COUNT(*) AS CHALLENGES_COUNT
```

```
        FROM HACKERS H1
        JOIN CHALLENGES C1
        ON H1.HACKER_ID = C1.HACKER_ID
        GROUP BY H1.HACKER_ID
    )
)
OR COUNT(*) IN (
    SELECT UNIQUE_CHALLENGES
    FROM (
        SELECT COUNT (*) AS UNIQUE_CHALLENGES
        FROM HACKERS H2
        JOIN CHALLENGES C2
        ON H2.HACKER_ID = C2.HACKER_ID
        GROUP BY H2.HACKER_ID
    )
    GROUP BY UNIQUE_CHALLENGES
    HAVING COUNT(*) = 1
)
ORDER BY CHALLENGES_CREATED DESC, H.HACKER_ID ASC;
```

Q2.

You did such a great job helping Julia with her last coding contest challenge that she wants you to work on this one, too!

The total score of a hacker is the sum of their maximum scores for all of the challenges. Write a query to print the hacker_id, name, and total score of the hackers ordered by the descending score. If more than one hacker achieved the same total score, then sort the result by ascending hacker_id. Exclude all hackers with a total score of 0 from your result.

Input Format

The following tables contain contest data:

Column	Type
hacker_id	Integer
name	String

- Hackers: The hacker_id is the id of the hacker, and name is the name of the hacker.
- Submissions: The submission_id is the id of the submission, hacker_id is the id of the hacker who made the submission, challenge_id is the id of

Column	Type
submission_id	Integer
hacker_id	Integer
challenge_id	Integer
score	Integer

the challenge for which the submission belongs to, and score is the score of the submission.

Solution:

```
SELECT HACKER_ID, NAME, SUM(MAX_SCORE) AS TOTAL_SCORE
FROM (
    SELECT H.HACKER_ID, H.NAME, S.CHALLENGE_ID, MAX(SCORE) AS
MAX_SCORE
    FROM HACKERS H
    JOIN SUBMISSIONS S
    ON H.HACKER_ID = S.HACKER_ID
    GROUP BY H.HACKER_ID, H.NAME, S.CHALLENGE_ID
)
GROUP BY HACKER_ID, NAME
HAVING SUM(MAX_SCORE) > 0
ORDER BY TOTAL_SCORE DESC, HACKER_ID ASC;
```